

BUSINESS REQUIREMENTS DOCUMENT

Customer 360 & Product Growth Intelligence Platform

Meesho-style social-commerce marketplace analytics: RTO reduction, funnel conversion and customer retention

Field	Detail
Document	Business Requirements Document (BRD)
Version / Status	1.0 / Draft for stakeholder review
Prepared by	Shardul (Business / Data Analyst)
Data scope	50,000 orders (70,032 line items), 10,000 customers, 500 SKUs; 1 Jan 2025 to 1 Jan 2026
Deliverables covered	SQL warehouse (meesho_growth_db), Python analytics, 7-page Power BI dashboard, AI Business Analyst assistant, Smart-Confirm product recommendation
Companion documents	Phases 13-17 documentation (Word), README.md, ai_assistant/app.py

1. Executive Summary

Meesho-style marketplaces grow on price-sensitive Tier-2/3 demand, but revenue is lost at two points: parcels that return undelivered (Return-to-Origin, RTO) and shoppers who abandon at checkout. Of Rs 42.29 Million gross revenue, only Rs 24.20 Million (57.2%) is delivered; Rs 6.08 Million (14.38%) is tied up in RTO.

This project delivers one governed analytics platform that quantifies these leaks, segments the 10,000-customer base, tracks cohort retention, and converts the findings into a prioritised product plan. The headline recommendation is **Smart-Confirm**, an automated WhatsApp COD verification and address-quality engine, targeting RTO of 11.0% within two quarters.

2. Business Problem & Opportunity

Problem	Evidence	Business impact
High RTO	RTO rate 14.38% (about 7,190 of 50,000 orders); COD cited as 68% of RTO (see Section 13)	Rs 6.08M revenue not realised; wasted freight (about Rs 0.72M at Rs 100 per order, assumed); locked inventory
Cart-to-checkout abandonment	Funnel 100k search, 65k filter, 38k cart, 22k checkout, 10k order; 42.1% drop from cart to checkout	Late shipping/handling fees on low-margin baskets; each 10-point reduction adds about 1,745 orders per 100k searches
Weak reseller conversion	21,191 WhatsApp share events in the data; share-to-purchase not yet attributed	Free acquisition channel under-monetised
Revenue realisation gap	Delivered 57.2% of gross; net realised revenue Rs 30.18M	Leadership lacks one trusted view of cash vs booked revenue

3. Project Scope

3.1 In scope

- Data warehouse and flat analytics model covering sales lines, orders, customers, products and session-level feature events.
- Data cleaning and standardisation (status values, dates, payment class, null handling).
- RFM customer segmentation, CLV bands, churn status and monthly cohort retention.
- Power BI dashboard (7 pages) with DAX measures for revenue, RTO, funnel, retention and growth.
- Natural-language AI Business Analyst assistant (command line) over meesho_growth_db.
- Product artifacts: decision framework, user stories, PRD (Smart-Confirm), RICE prioritisation, 3-quarter roadmap, QA checklist.

3.2 Out of scope

- Changes to courier-partner systems, payment gateway or the live transactional platform.
- Real-time (streaming) analytics; the platform is batch/refresh based.
- Production build of Smart-Confirm; this project specifies and prioritises it.
- Machine-learning RTO prediction (roadmap item for Q3).

4. Business Objectives & Success Metrics

#	Objective	Baseline	Target	Timeframe
O1	Reduce overall RTO rate	14.38%	11.0%	2 quarters
O2	Reduce cart-to-checkout drop-off	42%	32%	2 quarters
O3	Confirm COD orders before dispatch	0%	70% or more	Q1 (after launch)

#	Objective	Baseline	Target	Timeframe
O4	Raise delivered-revenue realisation	57.2%	Track monthly; +1 pt = about Rs 0.42M	Ongoing
O5	Lift WhatsApp share-to-purchase conversion	3.0% (assumed)	4.5%	2 quarters
O6	Single governed dashboard adopted by all four stakeholder groups	Fragmented reports	One dashboard, signed-off KPI definitions	Project close

5. Stakeholder Matrix

Stakeholder	Needs	Key indicators
CEO	One trusted view of growth, cash conversion and risk; clear investment priorities	Gross vs delivered revenue, realisation rate, RTO rate, profit margin
Marketing Head	Which channels and segments bring repeat, profitable buyers; targeted win-back	Channel mix, Champions vs At Risk, M1 retention, CLV band, WhatsApp shares
Product Lead	Locate funnel leaks; prioritise features on evidence	Stage conversion, 42% cart drop, feature usage, rating and returns by SKU
Logistics Lead	Fewer failed deliveries, lower reverse freight, risky-pincode flags	RTO by payment class, state, city, pincode; COD share; stock cover

6. In-Scope Business Questions

#	Category	Business question
1	Customer 360	How many customers sit in each RFM segment; how large is Champions vs At Risk?
2	Customer 360	What share of customers is repeat vs one-time, and what is their lifetime value?
3	Customer 360	Which acquisition channels produce the highest-CLV, lowest-churn customers?
4	Customer 360	What is cohort retention at Month 1, 2 and 11; which cohorts retain best?
5	Revenue & RTO	What are gross, delivered and net realised revenue; where is the gap?
6	Revenue & RTO	What is the RTO rate and revenue lost, overall and by state, city, pincode?
7	Revenue & RTO	How do COD and Prepaid orders compare on RTO rate and value?
8	Revenue & RTO	Which categories and discount bands are most and least profitable?
9	Product Funnel	What is the stage-by-stage conversion from search to completed order?
10	Product Funnel	Where is the biggest drop-off; which segments or devices drive it?
11	Product Funnel	Which SKUs have high return rates or stockout risk?
12	Product Funnel	How do ratings and discounts relate to conversion and returns?
13	Growth	Which cities are the top 10 high-growth markets?
14	Growth	How effective is WhatsApp sharing at driving purchases?
15	Growth	What revenue is at risk from churn; how much could win-back recover?

7. Business Requirements

Priority uses MoSCoW (Must / Should / Could). 'Traces to' links each requirement to the business questions in Section 6.

ID	Requirement	Priority	Traces to
BR-01	The platform shall report gross, delivered, shipped, cancelled, RTO and net realised revenue, with a reconciliation that sums to gross revenue.	Must	Q5
BR-02	The platform shall report RTO rate and RTO revenue, filterable by payment class, state, city and pincode.	Must	Q6, Q7
BR-03	The platform shall assign every customer an RFM segment, CLV band and churn status.	Must	Q1, Q2, Q15
BR-04	The platform shall show monthly cohort retention (month index 0-12) with pooled KPIs for M1, M11 and M12; months not yet complete shall be excluded or shown as N/A, never as blank or zero.	Must	Q4
BR-05	The platform shall show the search-to-order funnel with stage conversion and drop-off percentages.	Must	Q9, Q10
BR-06	The platform shall report gross profit and margin by category and discount band.	Should	Q8
BR-07	The platform shall identify SKUs with high return rates or low stock cover.	Should	Q11, Q12
BR-08	The platform shall rank cities by growth and show regional RTO.	Should	Q6, Q13
BR-09	The platform shall measure WhatsApp share volume and, once links are tagged, share-to-purchase conversion.	Should	Q14
BR-10	Business users shall be able to ask plain-English questions and receive a finding, context and recommendation (AI Business Analyst).	Should	Q1, Q5, Q6, Q8, Q9
BR-11	All KPI definitions shall be documented and reconcile across SQL, Python and DAX.	Must	All
BR-12	The dashboard shall respond correctly to slicers and cross-highlighting on every page.	Must	All

8. Data Requirements

Table	Grain	Key fields	Used for
meesho_sales_flat_analytics	Order line (70,032 rows)	Order_ID, Customer_ID, SKU, Order_Date, Revenue, Status, Payment_Method, Units_Sold, prices, Discount_%	Revenue, RTO, profit, cohorts
meesho_orders	Order (50,000 rows)	Order_ID, Customer_ID, Total_Amount, Payment_Method, Status, Pincode	Order-level checks
meesho_customers	Customer (10,000 rows)	Customer_ID, RFM Segment, CLV Band, Churn Status, First Order Cohort, Acquisition_Channel	Customer 360, retention
meesho_products	SKU (500 rows)	SKU, Product_Category, Supplier_Price, Base_Price, Product_Rating, Stock_Qty	Category profit, stock
meesho_feature_events	Event (150,000 rows)	Session_ID, Customer_ID, Feature_Name, Converted_Flag, Timestamp	Funnel, WhatsApp shares

Status vocabulary: DELIVERED, SHIPPED, CANCELLED, RTO_COMPLETE. **Payment class:** COD or Prepaid (derived from Payment_Method).

9. Customer Process Flow (Current vs Proposed)

Step	Stage	What happens	Volume / risk
1	Guest Visit	Shopper opens the app or a shared WhatsApp link	Session starts
2	Search / Filter	Search bar or category filter used	100k searches; 65k use a filter
3	Product Detail View	Price, rating and reviews viewed	Proposed: pincode delivery estimate and fee shown here
4	Cart Addition	Item added to cart or wishlist	38k sessions; 42% drop before checkout (fees shown late)
5	COD / UPI Selection	Shopper picks payment method	22k checkout clicks; UPI is lowest RTO risk
6	Proposed: Smart-Confirm	COD orders get an Address Quality Index; WhatsApp confirm (YES / CANCEL / EDIT)	No reply in 24h or CANCEL: order cancelled, stock released
7	Order Dispatch	Confirmed orders handed to courier	10k completed orders
8	Delivery or RTO Return	Parcel delivered, or refused / unreachable and returned	Delivered: revenue realised. RTO: revenue lost (14.38%)

10. Assumptions, Constraints & Dependencies

Type	Item
Assumption	Reverse plus forward freight is Rs 100 per RTO order (to be replaced with actual logistics cost).
Assumption	WhatsApp share-to-purchase baseline of 3.0% until share links carry attribution.
Assumption	Dataset (Jan 2025 to Jan 2026) is representative of ongoing behaviour.
Constraint	Month-12 retention is unavailable until a cohort has 12 complete months (data ends 1 Jan 2026).
Constraint	Batch refresh only; no real-time data.
Dependency	WhatsApp Business API access and message-template approval for Smart-Confirm.
Dependency	Pincode serviceability master and courier delivery-time data for delivery estimates.

11. Risks & Mitigations

Risk	Likelihood	Impact	Mitigation
Stricter COD verification lowers COD conversion	Medium	High	A/B holdout; tune AQI thresholds; track order value and RTO together
Source numbers disagree with documented figures (Section 13)	High	High	Re-derive from source data and update the document before publishing
WhatsApp API downtime or opt-outs	Low	Medium	SMS and in-app fallback; retry with back-off; manual review queue
Planning assumptions prove wrong	Medium	Medium	Tag assumptions; replace with measured values after first sprint
Low dashboard adoption	Medium	Medium	Stakeholder walkthroughs; signed-off KPI dictionary

12. Product Recommendation Summary

Features were scored with $RICE = (Reach \times Impact \times Confidence) / Effort$ (planning estimates).

Rank	Feature	Reach	Impact	Conf.	Effort	RICE
1	WhatsApp COD Verification Bot	6,800	3	80%	4	4,080.0
2	Pre-Checkout Pincode Delivery Estimate	3,800	2	80%	2	3,040.0
3	One-Click Reordering for Repeat Buyers	9,579	0.5	80%	1.5	2,554.4
4	Gamified Instant UPI Discount	2,200	2	50%	3	733.3
5	Reseller Social Sharing Leaderboard	1,500	1	50%	3	250.0

13. Data Reconciliation Notes (Open Issues)

Headline figures were checked against the Power BI model. Gross revenue (Rs 42.29M), delivered revenue (Rs 24.20M), RTO revenue (Rs 6.08M) and RTO rate (14.38%) match. Two documented figures do not yet reconcile:

Issue 1 - COD share of RTO. The model shows COD is 19.8% of orders and about 20% of RTO revenue (Rs 1.23M of Rs 6.08M), with a COD RTO rate (14.5%) close to Prepaid (14.4%). The 68% figure used in this BRD must be re-derived from source data. If the model is correct, the case for COD-specific intervention weakens and address quality and delivery-fee transparency become the stronger levers.

Issue 2 - Funnel counts. The feature_events table holds about 21,000 events per feature and does not form a sequential funnel. The 100k to 10k funnel is used as the documented baseline and should be rebuilt from session-ordered events.

14. Acceptance Criteria

#	Criterion
1	Revenue by status (Delivered + Shipped + Cancelled + RTO) sums exactly to gross revenue (Rs 42.29M) in SQL, Python and Power BI.
2	RTO rate reconciles to 14.38% in all three layers.
3	Retention cards return M1 about 33.1%, M11 about 34.4%, M12 'N/A' and overall about 34.1% with no blank values.
4	All 15 business questions can be answered from the dashboard or the AI assistant.
5	Every slicer and cross-highlight behaves correctly on all seven pages.
6	Issues 1 and 2 in Section 13 are resolved or formally accepted by the Product Lead.

15. Approval & Sign-off

Role	Name	Signature	Date
CEO			
Marketing Head			
Product Lead			
Logistics Lead			
Business Analyst	Shardul		